

Field-Tested Authentication for Quantum Key Distribution and DoS Attacks

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Abstract. Authentication is a crucial requirement for the security of Quantum Key Distribution (QKD). Yet, the integration of suitable methods in QKD systems tends to receive little attention from the research community. As a result, Wegman-Carter message authentication established itself as the go-to solution, leading to serious inefficiencies and additional trust assumptions, making it hard to recover from denial-of-service attacks. Another method is to use the lattice-based signature scheme Dilithium, as proposed by Wang et al. [36]. This method avoids the drawbacks of Wegman-Carter but, unfortunately, introduces new disadvantages. In this work, we implement and test several authentication methods on an actual QKD system. We compare and analyze three authentication variants, i.e., Wegman-Carter, Dilithium, and the established message-authentication code Chaskey, as a new method for authentication in QKD, which uses fewer quantum keys. We focus on the key consumptions, runtimes, and practicality in a field test of the QKD system. Lastly, we take a broader look at authentication for QKD in the context of Denial-of-Service attacks and propose a solution by combining several authentication methods to achieve their individual advantages while simultaneously avoiding several drawbacks.

Keywords: QKD, Authentication, Post-Quantum Security

1 Introduction

Cryptography is an indispensable tool to secure communication. A core concept of cryptography—and historically the first example—is the encryption of

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messages between two parties Alice and Bob that share some key (symmetric encryption). In modern cryptography, keys for symmetric encryption are a sequence of random bits. A core challenge is how Alice and Bob can exchange such a key in a secure manner.

In their seminal work [10], Diffie and Hellman introduced the concept of asymmetric encryption where Alice and Bob do not need to share a key ahead of time. Due to efficiency reasons, however, asymmetric encryption is mainly used to securely exchange symmetric keys (the dedicated primitive is called key-encapsulation mechanism) which is then used to encrypt the actual messages. Constructions for asymmetric encryption based on number-theoretic problems like factoring and discrete logarithms that are used nowadays are known to be insecure once large-scale quantum computers exist, due to Shor’s algorithm [33], which shows the devastating effects that quantum computers have on public-key cryptography. An alternative is post-quantum cryptography that relies on other hardness assumptions, e.g., the hardness of finding shortest vectors in lattices, which is assumed to be resistant to attacks with quantum computers. However, there is the possibility that an efficient quantum—or even classical—algorithm invalidating these assumptions will be found in the future. Luckily, quantum mechanics also offers methods to foster new promising candidates to securely exchange a secret between two parties: *Quantum Key Distribution* (QKD), introduced by Bennett and Brassard [5] forty years ago, indeed presents a method to harness the fundamental rules of quantum mechanics to exchange keys between two distant parties. Although QKD promises the exchange of information-theoretically secure keys thanks to quantum uncertainty (no-cloning theorem [38, 39]) and verifiable entanglement, the practical application of this idea is faced with a couple of fundamental challenges. Among those—and the one of interest for this paper—is the problem of authenticating the classical communication needed between Alice and Bob, to ensure Eve cannot temper with the protocol’s execution.

A typical method for authentication is to use the Wegman-Carter message authentication code (MAC) as mentioned in several works [8, 20, 34, 37]. Here, messages are authenticated using one-time keys. A clear advantage of the Wegman-Carter authentication is that it achieves information-theoretic security. The disadvantage is that Alice and Bob need some pre-shared key to authenticate the first quantum key distribution session. A more recent approach [36] suggests to use Dilithium, a lattice-based signature scheme, for the authentication. While this avoids the problem of requiring a pre-shared key, the security now relies on computational hardness assumptions, more precisely, lattice assumptions.

An often neglected problem of QKD are Denial-of-Service (DoS) attacks. While Alice and Bob are able to determine whether an adversary was eavesdropping on the quantum channel, they cannot prevent the adversary from doing so—being able to identify such an eavesdropping adversary allows Alice and Bob to discard any potential key bits that the adversary might have learned. When using Wegman-Carter for authentication, every attempt to exchange a new quantum key—regardless of being successful or unsuccessful due to an eavesdropping

adversary—consumes part of the Wegman-Carter key. By constantly eavesdropping on the quantum channel for a sufficiently long time, the adversary can “force” Alice and Bob to use all key material that they share.⁴

1.1 Contribution

In this work, we address the aforementioned issues. Besides the presented authentication methods, i.e., Wegman-Carter (mentioned in several works [8, 20, 34, 37]) and Dilithium (considered in [36], and recently standardized by NIST as ML-DSA [30]), we additionally consider authentication via the message authentication code Chaskey [28] which has not been considered for QKD yet. In contrast to other MACs, Chaskey indeed comes with an explicit security proof against quantum adversaries [2], while it does not constantly consume one-time-pad keys, making it more interesting in the context of QKD.

We provide practical results for the three different authentication approaches recorded on the field-tested QKD network presented in [13]. Hereby, we demonstrate the operability of the authentication schemes in a real QKD system, investigating runtimes and key consumptions. While several QKD field tests mention the utilized authentication variant, the exact choice of parameters tends to get very little attention in most present works. Here, we take a detailed look into parameter choices, especially for the Wegman-Carter scheme, making impacts of the choice of security parameter and size of classical channel communication clear. Furthermore, we showcase that our implementations can handle the amount of data to be authenticated in the system without delaying the QKD operation.

Going further, we propose a combination of the three different authentication methods, Wegman-Carter, Dilithium, and Chaskey, to obtain their advantages while simultaneously avoiding many of their respective disadvantages. The initial QKD round (i.e., during the first key exchange) will be authenticated using Dilithium. From then on, authentication will be done by Wegman-Carter and Chaskey: The former is the default option, while the latter is used in case the key material for Wegman-Carter is depleted—for instance, due to a DoS attack—or in low QKD key rate scenarios where authentication would consume most of the newly generated key.

One advantage of our approach compared to [36] is that lattice assumptions are only required to hold for the very first round. Also, by ensuring that Alice and Bob always have a key for Chaskey as a backup, we limit the vulnerability to DoS attacks implied by a Wegman-Carter-only authentication approach, such as [17]. This, however, comes at the cost of assuming a random permutation used to instantiate Chaskey. Compared to lattice assumptions (or more generally post-quantum hardness assumptions), this is less critical as one can simply replace permutations if they turn out to be insufficient or their security becomes doubtful.

⁴ Strictly speaking, Alice and Bob could refrain from using the system but they will not be able to determine a priori if an attempted QKD session will be successful or fail due to an eavesdropping adversary.

1.2 Related Work

The problem of authenticated QKD post-processing is far from being new. The first works that mention the need for an incorruptible public channel can be traced back as far as 1992, by Bennett et al. [4], where it was immediately suggested that Wegman-Carter [37] was a promising candidate to solve this issue.

Using Wegman-Carter or other Information-Theoretic secure (ITS) message authentication codes [1, 18] indeed perfectly fits in with the goal of achieving unconditionally secure key exchange thanks to QKD. As such, Wegman-Carter remained the default go-to authentication method for QKD for numerous years. This method, however, uses a significant portion of the QKD key across its execution. In some high-loss, hence low-key-rate scenarios [22], this could cause no QKD key to be produced at all. Thus, many papers are working on optimising the key rate/key consumption ratio. While some of those works aimed to improve this ratio and retain the information theoretical security of the authentication (and thus of the QKD protocol) [17, 25], we also saw the appearance of new protocols trading the ITS property for better performances [7, 32].

Lately, more researchers voiced their concerns about ITS MACs' performances on key consumption and bootstrapping of QKD networks [19, 41]. While many researchers still consider ITS to be the core property of interest of QKD [25], those concerns lead to significant works studying the incorporation of non-ITS MACs and post-quantum cryptography to QKD authentication [7, 35, 36]. A common argument to those works, to which we also stand by in this paper, is that the core property of interest of QKD is the unconditional forward-secrecy it provides (i.e., secrecy of the unused QKD key persists in time, independently of the evolution of Eve's computational power — also called everlasting security [7]), as long as we can provably authenticate the post-processing at a given time t .

While we, in this work, exclusively focus on classical authentication methods, it is important to note that some purely quantum-based authentication methods have been developed (quantum identity authentication [11]). Another authentication variant that maintains ITS properties are physically unclonable functions (PUFs) [31]. However, they require an addition to the physical setup and pre-sharing of a database (essentially containing the challenge-response pairs of the PUFs). While it may be interesting to compare their performances with classical techniques, this is out of scope for this paper.

1.3 Structure of this Work

The rest of this work is structured as follows. First, we introduce preliminaries in Section 2 on the QKD system used for testing and on the principles of Wegman-Carter, Dilithium and Chaskey authentication schemes. Section 3 contains the field test of the authentication schemes on the QKD system, including preparations and results. Section 4 concludes this work with the proposal of a combined protocol which is more resilient to DoS attacks on QKD systems.

2 Preliminaries

To implement a QKD system, a quantum channel and a classical channel are required [15,40]. The purpose of the quantum channel is to serve as a distribution platform for the quantum states between the two involved parties that want to exchange a symmetric key. For the quantum channel, photons are usually distributed via glass fiber or free-space optical links (e.g., satellite-to-ground QKD) [15,21,24] and are encoded in two non-orthogonal bases to leverage the no-cloning theorem of quantum mechanics. There are many different protocols to implement the quantum channel for QKD.

The QKD setup [13] we use across this paper implements an entanglement-based variant of the BBM92 protocol. Here, a central photon pair source distributes photon pairs to Alice and Bob, who both act as receivers. The source does not yield information on the secret key [6] and thus can be considered untrusted (i.e., controlled by the adversary). The exact implementation of the quantum channel and protocol is beyond the scope of this work. For a more detailed explanation, we refer the interested reader to [13].

The classical channel is an authenticated communication channel that serves as a data carrier for all information that Alice and Bob have to exchange during the protocol to extract a shared secret key from the quantum states [40]. Therefore, Alice and Bob perform the so-called post-processing phase of the protocol, which consists of several individual steps, including key sifting, error correction, error verification, and privacy amplification [40]. These post-processing steps are required to ensure that the resulting keys are indeed correct (identical) and secure (that means Eve's information on the shared key is minimized).

Notably, the authentication of this channel is necessary to prevent man-in-the-middle attacks on the QKD systems. However, the channel can be public, so no encryption is required.

In the QKD setup used for this work, Alice and Bob start with a calibration phase of the experimental setup to minimize errors on the quantum channel. Following that, they proceed with the actual key exchange by sending and measuring the quantum states via the quantum channel. For each QKD round, Alice and Bob record data for 90 s and afterwards immediately start evaluation (post-processing) of the recorded round, while at the same time, data acquisition for the next round is started. The post-processing via the classical channel employs the following steps:

The measured quantum states have bit values (0 and 1) assigned. Both parties now have individual bit arrays of 0's and 1's, which are referred to as raw keys. Upon completion of this process, key sifting takes place. Therefore, Alice and Bob bilaterally announce the bases they measured in but not the measurement outcome. They neglect all events where the bases did not match or only one party recorded an event. Consequently, they obtain a sifted key of the same length n .

However, Alice's and Bob's keys can still differ for a number of bits e , either due to imperfections of the measurement setup or the presence of an eavesdrop-

per, Eve, attacking the quantum channel, for instance, intercepting and resending information on the quantum channel. The quantum bit error rate (QBER) q for a key is therefore defined as $q = \frac{e}{n}$. The errors on the remaining sifted key bits are corrected through classical error correction algorithms, i.e., low-density parity-check codes (LDPC) [12], or, for our demonstration, Cascade [27], which iteratively computes, sends, and compares block-wise parities of the key between Alice and Bob. Upon completion of the error correction, a key verification takes place to confirm whether Alice and Bob successfully generated an identical key.

As the final post-processing step, privacy amplification takes place. Hereby, both parties account for leaked information (e.g., through QBER, information published during post-processing) and shorten the error-corrected key by means of a two-universal hash function such that Eve’s information on the key is minimized. As a result, Alice and Bob obtain a secure key whose length is shorter than the error-corrected key.

For communication during post-processing, a reliable authentication method must be implemented. Each of the listed steps requires the sending and receiving of messages of varying sizes, which can influence the parameters for authentication, as discussed in Section 3.1. Thus, it is especially important to customize authentication to the underlying QKD system.

Candidates to solve the authentication problem in QKD’s post-processing can be found in both asymmetric (signature schemes) and symmetric (MAC) cryptography. Of the three primitives we consider in this paper, the first one, Dilithium [26] is a post-quantum signature scheme that thus requires a public-key infrastructure (PKI). While this means it does not achieve information-theoretic security, it does allow for starting a QKD session without any a priori shared secret. The two other schemes we consider, Wegman-Carter [17, 37] and Chaskey [28], are MACs and thus use a fraction of the QKD key for authentication. The main difference between those two symmetric schemes lies in their security level (information-theoretic security vs. post-quantum security) and their key consumption. Indeed, as will be pointed out in Section 3.1, the one-time key approach of Wegman-Carter consumes significantly more key material than Chaskey.

In the following Sections 2.1, 2.2, and 2.3, we revisit the basics of each of the three authentication variants.

2.1 Wegman-Carter

Wegman-Carter [37] is an information-theoretically secure MAC and is commonly suggested for the authentication of QKD systems [17]. In 1981, Wegman and Carter [37] indeed showed how to obtain information-theoretically secure MACs from strongly-universal₂ hash functions. That is, hash functions belonging to a set of functions H from M (essages) to T (ags) which have the following property: for any $m_1, m_2 \in M$ with $m_1 \neq m_2$ and any $t_1, t_2 \in T$, there are $\frac{|H|}{|T|^2}$ functions in H which map m_i to t_i (for $i \in \{1, 2\}$). In particular, given sets M' and T' of messages and tags of respective sizes $s_{m'}$ and $s_{t'}$, the MAC will map

block sizes of $2s$ to strings of size s , with $s = s_{t'} + \log_2 \log_2(s_{m'})$. To do so, a sequence $f_1, \dots, f_n$ (for $n = \log_2 s_{m'} - \log_2 s_{t'}$) is selected from H such that all $f_x(m)$ are equal for a given $m \in M$. The tag t' for a given message m' is then computed as follows: m' is split in chunks of sizes s on which f_1 is applied before concatenating the results, which is processed similarly using f_2 , etc. After applying f_n remains a final string of size s , of which the $s_{t'}$ lowest bits are used as final tag. The key size needed to create a MAC for a $2sn$ bits long message is then of $4s \log_2(m')$ bits, which they claim is close to optimal [17].

While we refer to it as Wegman-Carter, the exact Protocol studied in this paper is based on the work by Kiktenko et al. [17], where they proposed modifications to the trivial use of Wegman-Carter to limit key consumptions when authenticating QKD rounds (cf. performances in Section 3.1). Their new protocol involves two main changes. First, building on an idea already sketched in the original paper [37], they use a combination of XOR operations and hash functions in order to reuse part of the key without sacrificing the security of the scheme. Then, they update the message flow to a delayed authentication scheme, so that only one message needs to get authenticated per round per direction. Additionally, based on this delayed authentication scheme, they implement a “Ping Pong” scheme, such that only one direction is authenticated per round and each way is authenticated in alternating rounds. This method halves the key consumption. However, we did not implement this latter optimisation, due to the structure of the QKD test system, and only mention it here for the sake of completeness.

The fact that Wegman-Carter does not rely on hardness assumptions, thus achieving information-theoretic security, is the biggest incentive to use it for QKD round authentication. However, while it perfectly fits within the chase for unconditional security of QKD protocols, this property does come with serious drawbacks. In particular, it cannot be used if Alice and Bob do not already possess some shared secret. In QKD networks consisting of more than two nodes, this would lead to a dramatic increase in the number of key pairs needed when adding a new party. This can be solved by using trusted relays, such as to perform the key exchange in a party-to-relay manner [14]. A downside of this approach is that all communication going through a relay can potentially be decrypted by it, which introduces quite a serious single point of failure. While the first issue is typically solved by factory pre-shared keys, this still leaves the door open for serious consequences in the case of denial-of-service attacks that would consume the preshared key. Indeed, exclusively using Wegman-Carter (and more generally any symmetric cryptographic scheme) does not offer any solution to exchange a key from nothing. Thus, this would force us to exchange a new secret out-of-band, which is rather impractical.

2.2 Dilithium

Dilithium [26] is a lattice-based signature scheme that was recently standardized by NIST as ML-DSA [30]. The scheme follows the Fiat-Shamir with aborts framework and its security is based on the learning with errors problem; we depict it in Algorithm 1.

Algorithm 1 Template for the Dilithium signature scheme from [26, Figure 1]

Gen(1^λ)

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1:  $\mathbf{A} \leftarrow R_q^{k \times l}$ 
2:  $(\mathbf{s}_1, \mathbf{s}_2) \leftarrow S_n^l \times S_n^k$ 
3:  $\mathbf{t} := \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ 
4: return  $(pk = (\mathbf{A}, \mathbf{t}), sk = ((\mathbf{A}, \mathbf{t}), \mathbf{s}_1, \mathbf{s}_2))$ 

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Sign(sk, M)

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5:  $\mathbf{z} := \perp$ 
6: while  $\mathbf{z} = \perp$  do
7:    $\mathbf{y} \leftarrow S_{\gamma_1 - 1}^l$ 
8:    $\mathbf{w}_1 := \text{HighBits}(\mathbf{A}\mathbf{y}, 2\gamma_2)$ 
9:    $c \in B_\tau := H(M \| \mathbf{w}_1)$ 
10:   $\mathbf{z} := \mathbf{y} + c\mathbf{s}_1$ 
11:  if  $\|\mathbf{z}\|_\infty \geq \gamma_1 - \beta$  or  $\|\text{LowBits}(\mathbf{A}\mathbf{y} - c\mathbf{s}_2, 2\gamma_2)\|_\infty \geq \gamma_2 - \beta$  then
12:     $\mathbf{z} := \perp$ 
13:  end if
14: end while
15: return  $\sigma = (\mathbf{z}, c)$ 

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Verify($pk, M, \sigma = (\mathbf{z}, c)$)

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16:  $\mathbf{w}'_1 := \text{HighBits}(\mathbf{A}\mathbf{z} - c\mathbf{t}, 2\gamma_2)$ 
17: return  $\llbracket \|\mathbf{z}\|_\infty < \gamma_1 - \beta \rrbracket$  and  $\llbracket c = H(M \| \mathbf{w}'_1) \rrbracket$ 

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Wang et al. [36] considered Dilithium for the authentication of a QKD system, aiming at growing its short-term⁵ post-quantum secure asymmetric keys in long-term (and arguably information-theoretically) secure QKD keys. In contrast with traditional QKD authentication methods, using a PKI eliminates the need for pre-shared symmetric keys. As a direct consequence, the previously mentioned scaling issue of Wegman-Carter is mitigated, as one only needs to store its own private key and a valid certificate in order to communicate with other nodes.

While solving the DoS issue mentioned earlier (cf. Section 2.1) and making it easier to work with QKD networks, using Dilithium presents a couple of drawbacks regarding performance and security. Traditionally, public-key cryptography indeed leads to higher signature sizes which, by its impact on the network, can be seen as a minor inconvenience. The security of Dilithium relies on lattice assumptions, which naturally poses two main issues. Most notably, the hardness assumption may end up being broken one day⁶, which poses the issue of both finding a new suitable signature scheme and then updating the system to the new scheme, which is usually not trivial to do [16, 23]. Moreover, reliance on a hardness assumption does not translate well in the context of a

⁵ We adopt this nomenclature from [36]. We stress that short-term secure keys are not to be confused with ephemeral keys.

⁶ Chen [9] recently presented an efficient quantum algorithm for a lattice assumption; though it was later found to be flawed.

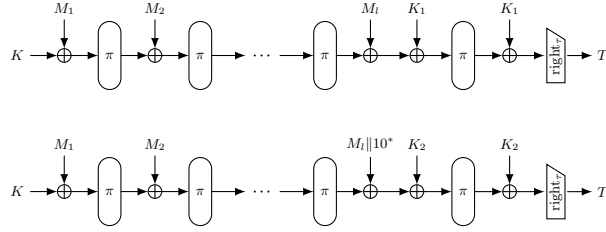


Fig. 1: Chaskey's mode of operation (taken from [28]) when $|M_l| = n$ (top), and when $0 \leq |M_l| < n$ (bottom). The subkeys K_1 and K_2 are generated from the initial key K , and $M_l || 10^*$ is shorthand for $M_l || 10^{n-|M_l|-1}$.

computationally unbounded adversary, against which researchers usually try to prove QKD secure.

2.3 Chaskey

Chaskey [28] is an ISO-standardized MAC which is visualized in Figure 1. Conceptually, Chaskey is very simple as it requires only a permutation and XOR operations.

To authenticate a message, Chaskey sets its key K as the initialisation vector for the first round of the computation's process. At round i , the message block m_i is first XORed with the input before applying Chaskey's permutation on the resulting string, and forwarding the output to the next round's input. For the last round, the last message block (padded if necessary) is XORed with the previous round's output, then XORed with a key K_j ($j \in \{1, 2\}$), derived from the key K . The result is then passed through the permutation again, before being once again XORed with the same K_j , resulting in the final tag.

Typically, Chaskey uses n -bit keys which also define the message blocks to be of length n . For $n = 128$, a key can be reused to authenticate up to 2^{48} n -bits message blocks (which, for $n = 128$, amounts to approximately 4 Petabytes of data). To prove security, Chaskey can be viewed as an instance of the (tweakable) Even-Mansour cipher, which recently resulted in an explicit security proof for Chaskey against quantum adversaries [2].

Chaskey provides an interesting middle ground between Wegman-Carter and Dilithium. Indeed, while it also belongs to symmetric cryptography and thus requires an a priori shared secret, its ability to authenticate many messages using the same key makes it practically invulnerable to DoS attacks⁷. However, Chaskey does rely on computational assumptions and therefore does not achieve information-theoretic security. Its security indeed relies on the hardness assumption that its underlying permutation is indistinguishable from a truly random

⁷ Strictly speaking, since there is an upper limit on how many messages can be authenticated with one key, an attacker against a QKD system could mount a DoS attack but due to the upper limit, it seems unrealistic that such an attack would succeed in practice.

permutation. While an efficient algorithm for lattice problems would have a devastating effect by breaking all lattice-based cryptographic primitives, one can always replace a broken permutation of Chaskey (or one for which security seems doubtful). Another advantage of Chaskey over Dilithium follows from the inherent differences between symmetric and asymmetric cryptography. Indeed, because we can use Chaskey together with a QKD (thus long-term, information-theoretically secure) key, we do not have to worry about offline attacks as long as we never used the key beforehand. This implies a key can be stored for an indefinite amount of time until we need to use it without compromising the protocol’s security. However, the key still needs to be stored securely.

3 Field-Test in a Live QKD-System

In the following, we elaborate on the implication of each scheme’s properties in the context of a real-world QKD system, briefly discuss the difference in criticality between the hardness assumption used for Dilithium and Chaskey, and present their respective performances evaluated on the QKD network in Sections 3.1 and 3.2.

3.1 Practical Implementation

Each of the schemes described in the previous section has been implemented and tested on the QKD infrastructure described in Section 2. In particular, we used Bouncy Castle’s implementation of Dilithium (that was at the time following NIST’s pre-standardisation recommendations), implemented Chaskey in Java according to a reference C implementation provided in [28] and implemented a new Wegman-Carter MAC in Java as well, following the variant by Kiktenko et al. [17], implementing a delayed authentication but without the ping-pong scheme.⁸

When testing, we used the parameter set for security level III for Dilithium as provided by the Bouncy Castle library, implemented 12 hashing rounds for Chaskey, as recommended by the ISO standard, and chose Wegman-Carter’s parameters as per Kiktenko et al.’s work [17], that is $\epsilon = 10^{-12}$ which, together with $n = 100$ allowed reuses, yields $\epsilon_{\text{eff}} = n\epsilon = 10^{-10}$. While Dilithium does not consume QKD generated key material for later authentication rounds, and Chaskey does only require an initial key, the Wegman-Carter scheme has a regular key consumption from each newly generated QKD round. This consumption is impacted by the choice of ϵ , as the required One-Time-Pad (OTP) key length per authenticated QKD round and direction (i.e., Alice $\rightarrow$ Bob and Bob $\rightarrow$ Alice)⁹ is given by [17]

$$\tau = \lfloor -\log_2 \epsilon \rfloor + 1 = 40 \text{ bits} . \quad (1)$$

⁸ Note, that we did not manage to obtain Kiktenko et al.’s original implementation and thus could not thoroughly test ours with respect to the original.

⁹ Note, that both directions of communication have to be authenticated independently. Hence, two OTP keys are required per QKD round.

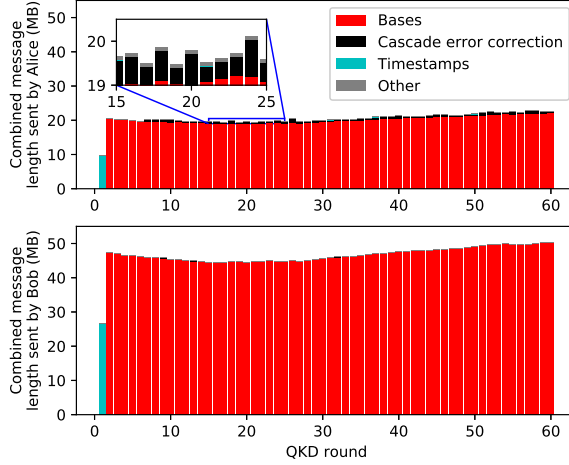


Fig. 2: Message sizes per QKD round via the classical channel between the two parties during key exchange. The first six rounds are for calibration of the setup (not authenticated). At round seven, the secure key generation starts. Most of the channel communication is caused by the basis exchange of the QKD measurements, a small amount by error-correction and a negligible amount by other communication. Alice and Bob sent different amounts of data as their respective photon detection rates differ.

Furthermore, the length of the recycled key L_{rec} depends on the maximum amount of data that has to be authenticated per QKD round per user μ . The recycled key is required to choose the applicable hash function. Hence, the key stays the same for several QKD rounds, before the hash function has to be refreshed to avoid disclosure and a new key is used (therefore the term recycled key). To choose an appropriate size for the recycled key, we have analyzed the data amount sent via the classical channel for a typical run of the QKD setup. The results of such a measurement are displayed in Figure 2. Notably, most of the channel communication is caused by the exchange of measurement bases of the photon detections. We note a large difference in the total data amount sent by Alice and Bob. The reason for this is different fiber lengths (Source $\rightarrow$ Alice and Source $\rightarrow$ Bob) causing different photon losses and hence a difference in detection rates. The slowly varying drift of the total data amount sent per round per party is caused by a small change in photon production by the source and can be neglected for our analysis. The first few rounds (in the presented measurement it is six rounds) serve for calibration of the setup. These calibra-

tion rounds serve for timing synchronization¹⁰ and to adjust the measurement devices such that in subsequent QKD rounds a low QBER can be achieved.¹¹ Therefore, data recorded during these rounds does not produce any secure key and hence, does not need to be authenticated. To extend on this argument, we discuss what happens, if Eve interferes with the system during the unauthenticated calibration rounds. Then, manipulation of the estimated QBER or timing offsets could occur, leading to an uncalibrated setup. However, if Alice and Bob proceed with the authenticated rounds to generate secure keys, while the setup is falsely calibrated, they will notice an increased QBER and therefore abort key generation and repeat the calibration. Hence, the worst-case scenario of Eve interfering during the unauthenticated calibration rounds would be a Denial-of-Service attack, which could be achieved by Eve anytime anyway, even during authenticated rounds. Therefore, not authenticating the calibration rounds saves QKD key consumption but does not reduce the security of the QKD protocol. For clarity, we do not display these calibration rounds in later figures. Note that the data amount sent each round can vary drastically in other QKD setups or when choosing different fiber lengths for the quantum channel. The results are also of importance for the use of Chaskey, as the expected data amount of one 90 s QKD round is much smaller than the maximum data size of ≈ 4 PiB. Hence, a very large number of QKD sessions will be possible with one key for Chaskey.

Following [17] for a typical data amount of 50 MB in the setup, we have to choose the length of the recycled key by

$$L_{\text{rec}} = 2\lambda\omega + \lambda + \tau + 1 = 417 \text{ bits}, \quad (2)$$

with $\omega = 31$ and λ being the smallest integer that fulfills

$$\left\lceil \frac{\mu}{\omega} \right\rceil^\lambda 2^{-\lambda\omega} \leq \epsilon - 2^{-\tau}. \quad (3)$$

From the definition of the OTP and recycled key lengths, we can calculate an average total key consumption per QKD round. As the recycled key has to be renewed after $n = 100$ uses, and both directions of communication between the two parties have to be authenticated, we get an average key consumption per QKD round c of

$$c = 2L_{\text{OTP}} + 2\frac{L_{\text{rec}}}{n} \approx 88 \text{ bits per QKD round} \quad (4)$$

for the used QKD setup when using the lightweight authentication scheme by Kiktenko et al. but without the ping-pong procedure. As one QKD round of the used setup takes 90 s, we obtain a key consumption of roughly 1 bit/s for

¹⁰ Alice and Bob are required to estimate their timing offsets for detections such that they can compare corresponding events.

¹¹ In the setup used here, the temperature of an interferometer placed at each receiver's site has to be adjusted. The requirement of this process arises from a two-photon Franson interference and therefore is beyond the scope of this work. It is explained in more detail in [13].

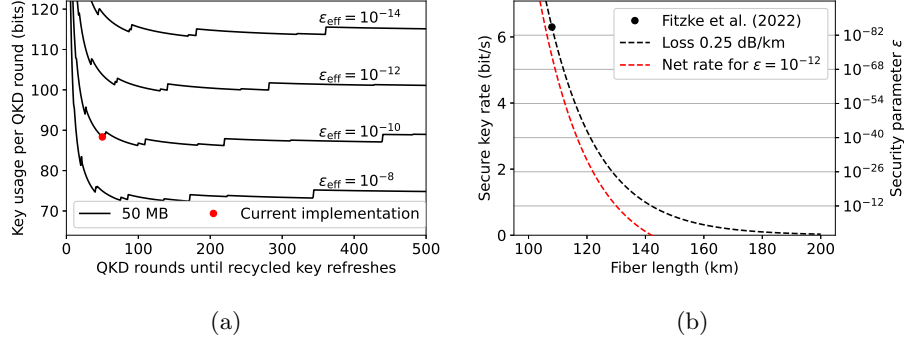


Fig. 3: (a) Total key consumption (recycled key + OTP key for both communication directions) for authentication with Wegman-Carter for various effective security parameters $\epsilon_{\text{eff}} = 2u\epsilon$, with u being the number of QKD rounds a recycled key remains valid. We assumed a maximum data amount of 50 MB per round to be authenticated per direction (Alice $\rightarrow$ Bob, Bob $\rightarrow$ Alice). (b) Secure key rate for various fiber lengths (black line), extrapolated from a previous QKD measurement via 108 km of fiber [13]. The right-hand side axis shows the maximum security parameter ϵ if the total generated key is used for authentication. In practice, one should choose the parameter such that a significant amount of QKD key is left for actual encryption. The red line shows the net key rate (=generated key minus authentication key) for an $\epsilon = 10^{-12}$.

$\epsilon = 10^{-12}$ and $\epsilon_{\text{eff}} = 10^{-10}$. The key consumption when using schemes based on the ideas of Wegman-Carter has to be considered for all QKD setups as it shrinks the effective rate a QKD system produces. This is especially important in high-loss and low-key-rate scenarios, as the authentication then consumes large chunks of the QKD keys. Hence, it is of particular interest to minimize the key consumption for a desired effective security parameter. Therefore, it might be desirable to change the length of the OTP key together with the recycled key and the number of rounds ($u = n/2$) it is reused. Therefore, we calculated $\epsilon = \epsilon_{\text{eff}}/n$ and subsequently the OTP and recycled key lengths following Equations (1) and (2). The results are depicted in Figure 3a, with our current implementation marked. While a small number of QKD rounds before refreshing of the recycled key drastically increases the key consumption, there is only a small change in key consumption for more than 100 QKD rounds. We note that for our effective security parameter of 10^{-10} , a value of $u = 99$ would be optimal. However, we opted against changing the parameters of our authentication, as the improvement would be marginal and for practical reasons when testing the QKD system.

In Figure 3b we made an exemplary calculation based on a QKD test over 108 km fiber published in [13]. The plot extrapolates how the key rate varies when

Table 1: Comparison of runtimes for generation of the authentication tag (Auth. time) and its verification (Verif. time) of the various algorithms. The test was run on an Intel i5-12400 with 64 GB RAM, Windows 10 Pro 22H2.

Algorithm	Data size (10^6 Bytes)	Auth. time (ms)	Verif. time (ms)
Dilithium-3 [26]	10	30	28
	25	69	69
	50	139	136
	100	280	274
	150	434	422
Chaskey-12 [28]	10	16	16
	25	40	40
	50	80	80
	100	160	160
	150	242	242
Wegman-Carter [17]	10	59	60
	25	149	152
	50	301	299
	100	598	601
	150	899	899

changing the fiber length by simply assuming a typical fiber loss of 0.25 dB/km ¹². On the right axis, we plotted the maximum security parameter ϵ that one could choose such that the complete secure key produced in a QKD round is consumed by the OTP keys for authentication. For a fixed $\epsilon = 10^{-12}$ we plotted the net key rate (difference between secure rate and OTP consumption for authentication). Clearly, in low key rate regimes, the key consumption for authentication drastically impacts the resulting key rate of the QKD system. Furthermore, this limits either the achievable quantum channel transmission distance or the choosable security parameter for authentication, thus impacting the practicality of such setups in real-world scenarios. Systems like the long-distance QKD experiment over 1000 km fiber [22] achieved a key rate for this distance of just 0.0034 bits per second, leaving basically no key for authentication via information-theoretically secure MACs. Hence, it is of interest to consider other options for authentication, such as Chaskey.

The runtime of an authentication scheme should be short to not delay the protocol execution and key generation. However, to this extent, it is not necessary to optimize for a few milliseconds of additional speed. For all three schemes (Dilithium, Wegman-Carter, Chaskey) we measured the times to compute the authentication tag and verify this tag for various amounts of arbitrary data. All

¹² For the sake of simplicity, we waived the impact of detector dark count rates on the key rate. Also note that in other regimes like short distances or high-detection rates, detector saturation effects have to be taken into account when calculating the key length.

show fairly similar capabilities and a linear time complexity with respect to the input data (cf. Table 1), with a slight edge to Chaskey in terms of overall speed.

3.2 Results

All three authentication methods have been tested in the QKD setup [13] with approximately 26 km of glass fiber for the quantum channel between Alice and Bob. If the algorithm requires a symmetric key at the beginning of the protocol (Chaskey and Wegman-Carter) it is taken from the key database of the system from an earlier QKD run. For Wegman-Carter, subsequent rounds repeatedly pull keys from earlier QKD rounds for the OTP key and renew the recycled key if necessary. For Dilithium no symmetric key from the key storage is required, however, we exchange the public keys of the involved parties before the actual QKD takes place. In particular, the results plotted in Figure 4 have been obtained by executing the QKD system for 60 rounds. The results do not display the rounds required for calibration as they are not authenticated and thus are of no interest for this work. All three implementations have been tested successfully, i.e., every QKD round that generated secure keys has been authenticated successfully. The variation of sifted key rates and QBERs, and consequently also secure key rates during the measurements, are caused by variations of the physical measurement setup (i.e., temperature fluctuations, drifting photon pair production rates at the source) and is thus independent of the software and authentication schemes in place. In fact, every measurement produced sifted key rates, averaging at roughly 150 bits/s and secure key rates of approximately 100 bits/s. For Wegman-Carter we have to deduct approximately 1 bit/s from the secure key rate as cost for authentication. The key consumption of Chaskey is negligible when conducting QKD over several rounds and maintaining the initial key, while Dilithium does not require a QKD key at all for authentication. The data size to be authenticated per QKD round was similar for all three runs. While Bob sent data of close to 50 MB per round, Alice sent approximately 20 MB. This is reflected in the authentication time of Bob being longer than authentication for Alice, and verification for Alice being longer than verification for Bob. However, in the used QKD system a delay much smaller than 1 s is still sufficiently short to not stop the QKD key generation with its blocks of 90 s. Nevertheless, as predicted by the runtime benchmark, we observe Chaskey to be the fastest algorithm. This property can make Chaskey an interesting option for QKD systems that have higher demands in terms of authentication speed such as high-key-rate systems.

In summary, we successfully tested three different means of authentication in a real QKD setup. While all variants proved to be sufficiently fast in the system, and are thus all suitable solutions for authentication, they all present different advantages. The biggest advantage of Wegman-Carter is its information-theoretic security, which comes with a few disadvantages, however. A minor downside is a small amount of secure key consumption ($\approx 1\%$), while the bigger downsides are the necessity of initial shared keys and its vulnerability to DoS attacks.

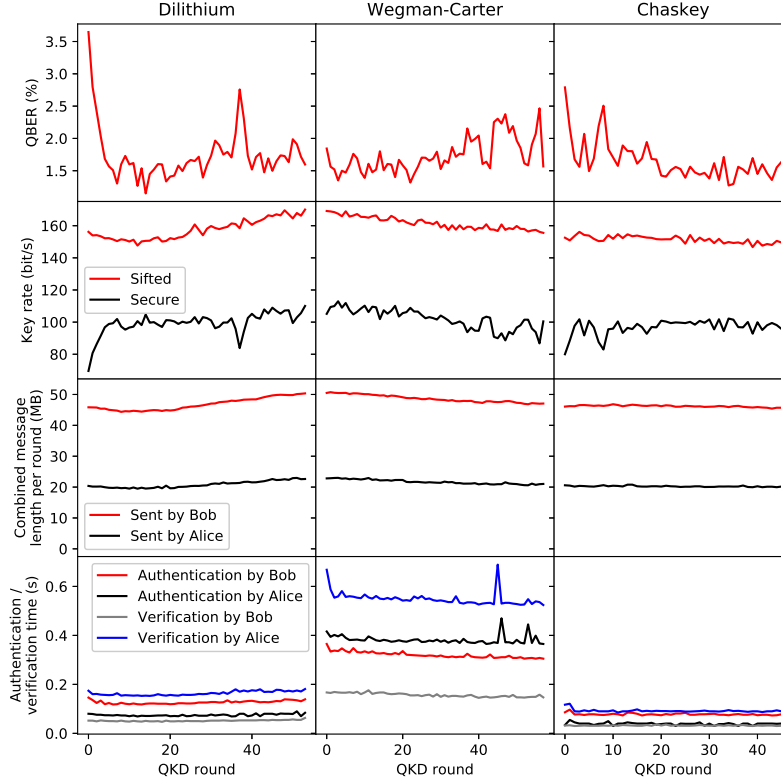


Fig. 4: Demonstration runs of the authentication algorithms on the live QKD-system. The calibration phases of the system before each measurement are not displayed here. QBER (quantum bit error rate) and key rates vary due to experimental reasons unrelated to the authentication. All QKD rounds were completed successfully without failing the authentication regardless of the used algorithms. The runtime advantage of Chaskey is clearly visible.

Dilithium offers post-quantum secure authentication via public-key infrastructures, thus not requiring the QKD keys for authentication. The obvious downside is the potential vulnerability due to advances in cryptanalysis against lattices. Chaskey as a new solution for QKD’s authentication has been successfully tested, offering negligible key consumption, easy-to-use implementation and fast computation times, making it a serious candidate for tailored QKD scenarios such as high-rate systems or high-loss links. It essentially avoids the downside of DoS attacks that Wegman-Carter has at the cost of giving up information-theoretic security, nevertheless maintaining the long-term security of the QKD key.

4 Combined Authentication Methods

While the performance of each scheme presented herein makes them all suitable for authenticating QKD rounds, they all come with their share of drawbacks. Their respective strengths and weaknesses, however, do seem to be complementary, drawing a picture for an efficient authentication protocol that would leverage all three schemes studied in this work. In the following sections, we sketch a new authentication protocol that uses Dilithium, Wegman-Carter, and Chaskey to compensate for each other’s weaknesses and attain a satisfactory practicality-security trade-off. We then briefly study the proposed solution’s security in the classical authenticated key-exchange framework, showing it retains unconditional forward secrecy.

4.1 Protocol Idea

The first step when creating a QKD network from scratch (or adding new nodes to the network) is the initial bootstrapping, meaning the establishment of an initial shared secret between the new node(s) and the already existing members of the network. As mentioned in Section 2.1, this is typically done through a combination of factory keys and trusted relays, which introduces a rather unsatisfactory single point of failure in the system. Instead of fully replacing Wegman-Carter by Dilithium, as done in [36], a natural idea would be to use Dilithium exclusively for the initial bootstrapping, such as to compensate for Wegman-Carter’s need for pre-shared keys. This shifts the trust to the underlying public-key infrastructure. While this does threaten the effective information-theoretic security of the scheme, we can play on the validity duration of emitted certificates to obtain a satisfyingly low adversarial success rate, such as not compromising the QKD key’s practical security.

When two parties communicate for the first time across the network, they would thus use the signature scheme (i.e., Dilithium) to authenticate the first QKD round, so they can extract a first symmetric key. From this point on, we rejoin the traditional QKD authentication method [17], i.e., every subsequent authentication is performed exclusively using this symmetric key (and those derived from it through a QKD execution), such as using the PKI as little as

possible. Ideally, those symmetric keys should be used together with Wegman-Carter (cf. Section 2.1), eliminating all concerns related to a potential break of some underlying computationally hard problem as is the case with PKI.

A persisting problem of only using Dilithium and Wegman-Carter, however, are DoS attacks. By performing such an attack, an adversary can make Alice and Bob use up the entire shared key from their storage—effectively taking them back to round one, where they would need to, again, rely on Dilithium to establish a shared key. To mitigate this issue, we introduce the additional use of Chaskey in the protocol. We impose the first QKD session to exchange not only a key for Wegman-Carter but also one for Chaskey. This additional secret is safely stored as long as the protocol succeeds using Wegman-Carter. In case the Wegman-Carter key gets fully used up, rather than using Dilithium, one would use the stored Chaskey key instead. As stated in Section 2.3, Chaskey’s key consumption is negligible, and any DoS attack aiming at using up its key would thus most likely fail, thus mitigating the chances that we would ever need to fall back to public-key cryptography. Switching back to Wegman-Carter authentication, once the QKD key storage is sufficiently filled up again, avoids relying on Chaskey’s security for the remaining time.

4.2 Security

Considering a computationally bounded adversary, we write ϵ_{QKD} , as the advantage of the adversary in guessing the key information exchanged in a QKD round, and ϵ_{WC} , ϵ_{C} , and ϵ_{D} , its advantage in breaking Wegman-Carter with key recycling [17, Definition 2], Chaskey [2, Theorem 7] and Dilithium [3], respectively. The final security parameter of any QKD protocol authenticated according to the proposed scheme can be upper bounded as per Eq. (5) [29].

$$\epsilon_{\text{full}} \leq \epsilon_{\text{env}} + N_{\text{WC}}(\epsilon_{\text{QKD}} + 2\epsilon_{\text{WC}}) + \delta_{\text{C}}(N_{\text{C}}\epsilon_{\text{QKD}} + \epsilon_{\text{C}}) \quad (5)$$

where $N = N_{\text{C}} + N_{\text{WC}}$ the total number of QKD rounds, with N_{C} (N_{WC}) the number of QKD rounds authenticated through Chaskey (Wegman-Carter); $\delta_{\text{C}} = 1$, if $N_{\text{C}} \neq 0$, and 0 otherwise. $\epsilon_{\text{env}} = \epsilon_{\text{D}} + \epsilon_{\text{Store}}$ encompasses the security of the key used in the first round, where ϵ_{Store} is the security parameter of its storage as meant in [17, Appendix B].

In particular, ϵ_{C} and ϵ_{D} only appear once (independently of how many rounds Chaskey and Dilithium authenticate) because their respective security bounds are computed on the whole lifespan of their key. In Chaskey’s case, given a 128-bit key, it means the bound holds for the authentication of up to 2^{48} bits (cf. Sec 2.3), which is well above the communication required for thousands of QKD rounds (cf. Sec 3.1). If the limit of 2^{48} bits is reached, another instance of Chaskey would be required. In Wegman-Carter’s case, the advantage is refreshed with each new tag. That is, it is refreshed each time we authenticate a message with it [17]. Because we have two authentications per round, it thus amounts to $2N_{\text{WC}} \cdot \epsilon_{\text{WC}}$ in the final bound.

Notice that, because we consider a computationally bounded adversary, we cannot claim information theoretic security anymore. However, as QKD authentication only needs to be secure at the time of the protocol’s execution, forward secrecy is ensured with respect to any adversary.

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A Data Availability

Our results build upon an existing QKD setup. The code to run the QKD setup is not part of this work and we can therefore not provide it. We can and will provide the data and code related to the authentication variants we present in this paper and the data extracted from the presented QKD measurements. Since this data is very large (many gigabytes per measurement), it is available upon request.